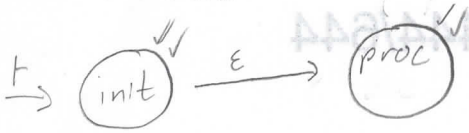
$\Sigma = \{1, 0, \#\}$ $\quad \Gamma / \Sigma = \{00, 01, 10, 11, \vdash, \sqcup, \#0, \#1, \underline{1}, \underline{0}, H, H1, H0\}$

always to right of R/W head

$25 + 271 - 1 = 295$



encoding: $0^{|Q|} 1\, 0^{|\Gamma|} 1\, 0^{|\Sigma|} 1\, 0^{\Phi^Q(s)} 1\, 0^{\Phi^Q(t)} 1\, 0^{\Phi^Q(r)} 1\, 0^{\Phi^\Gamma(\vdash)} 1\, 0^{\Phi^\Gamma(\sqcup)} 1$

$0^{\Phi^Q(q)} 1\, 0^{\Phi^\Gamma(\gamma)} 1\, 0^{\Phi^Q(q')} 1\, 0^{\Phi^\Gamma(\gamma')} 1i \qquad R=1, L=0$

✓ move to #:
moves the 1 to the hash w/ read head just after #1

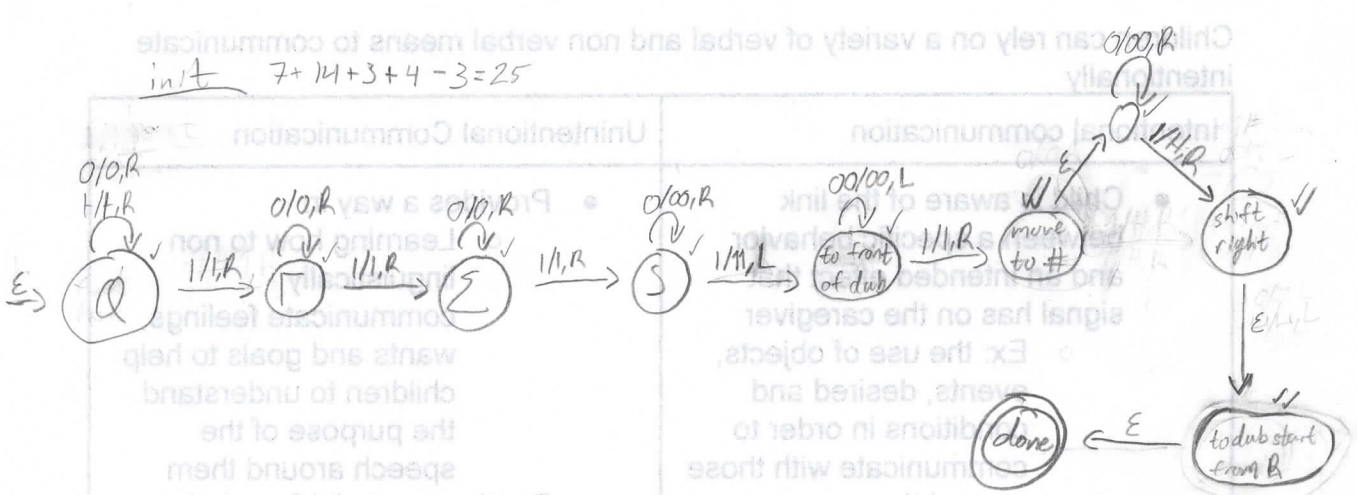
✓ shift right:
moves 1s & 0s one to right until ⊔

✓ to dub start from R:
puts read head just to left of dub, assuming we're R of dub

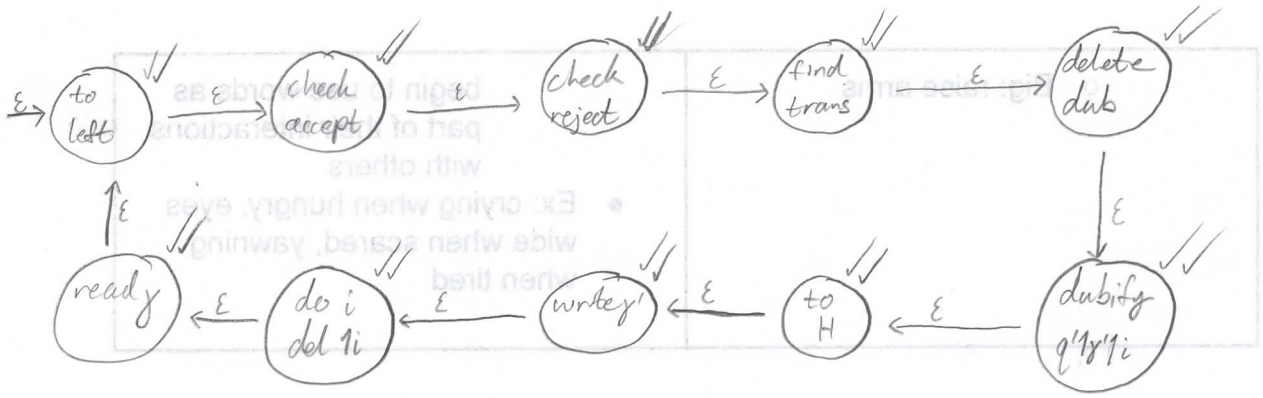
✓ shift dub L:
shifts dub one to the left, keeping the read head just to left

✓ to left:
shifts dub left (shift dub L) until reading ⊢

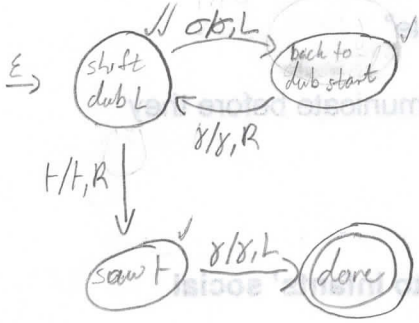
init $\quad 7 + 14 + 3 + 4 - 3 = 25$



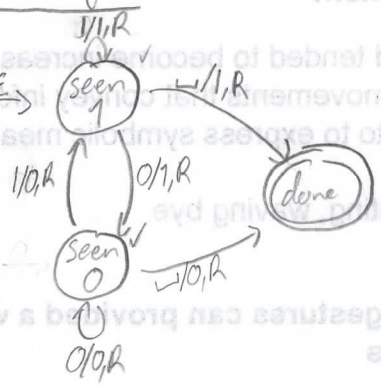
proc 12+39+12+75+3+6+15+11+76+33−10 = 271



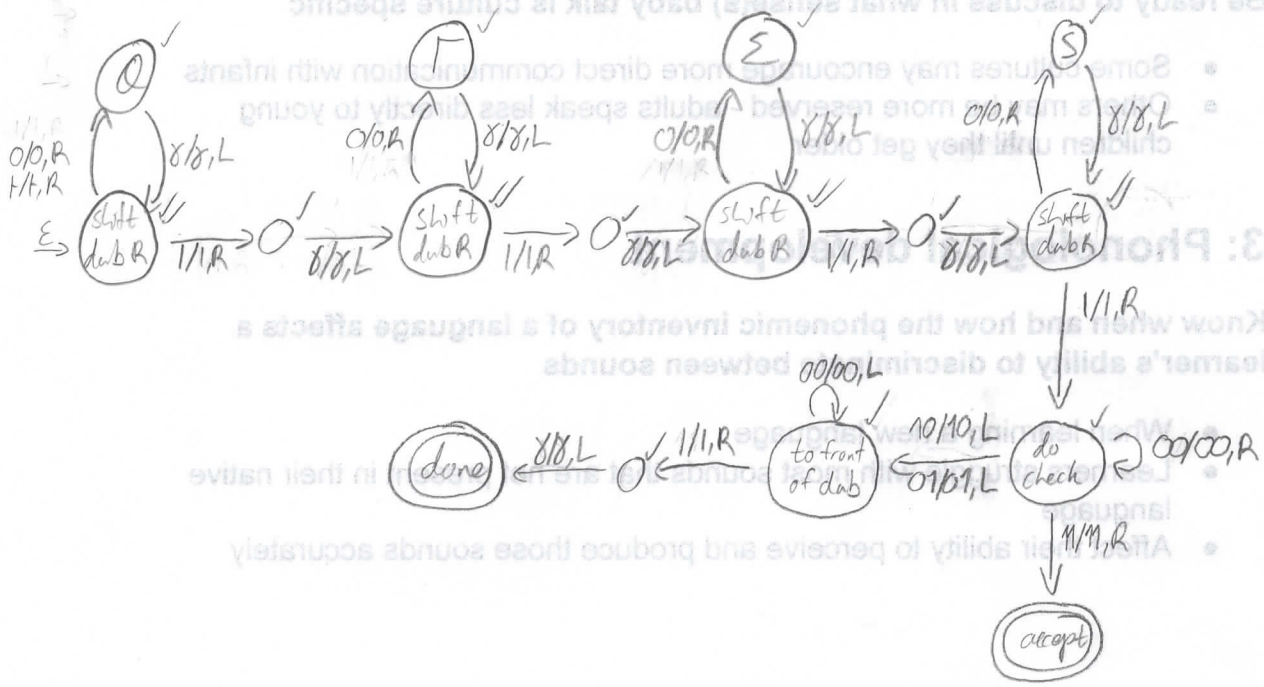
to left 3+9=12



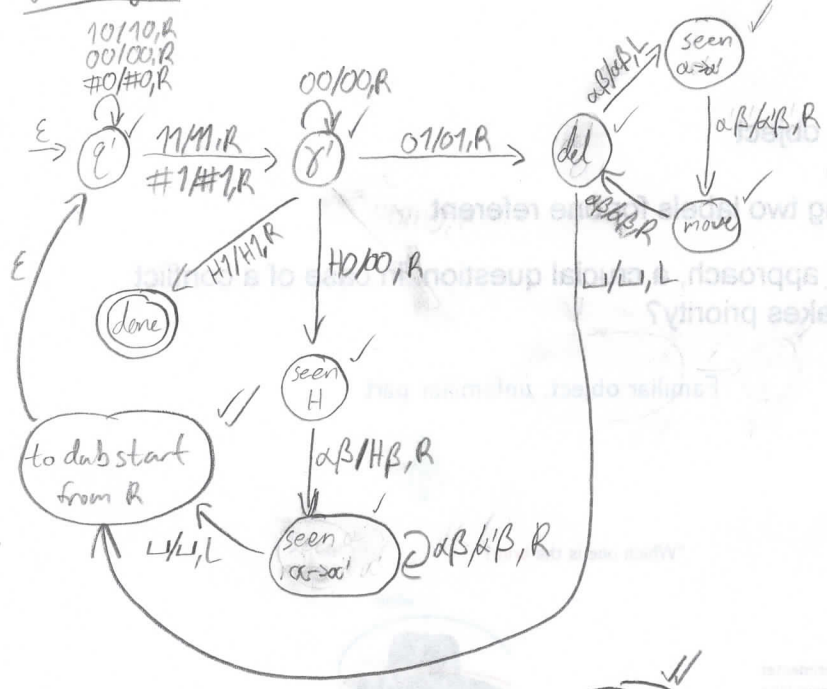
shift right 3



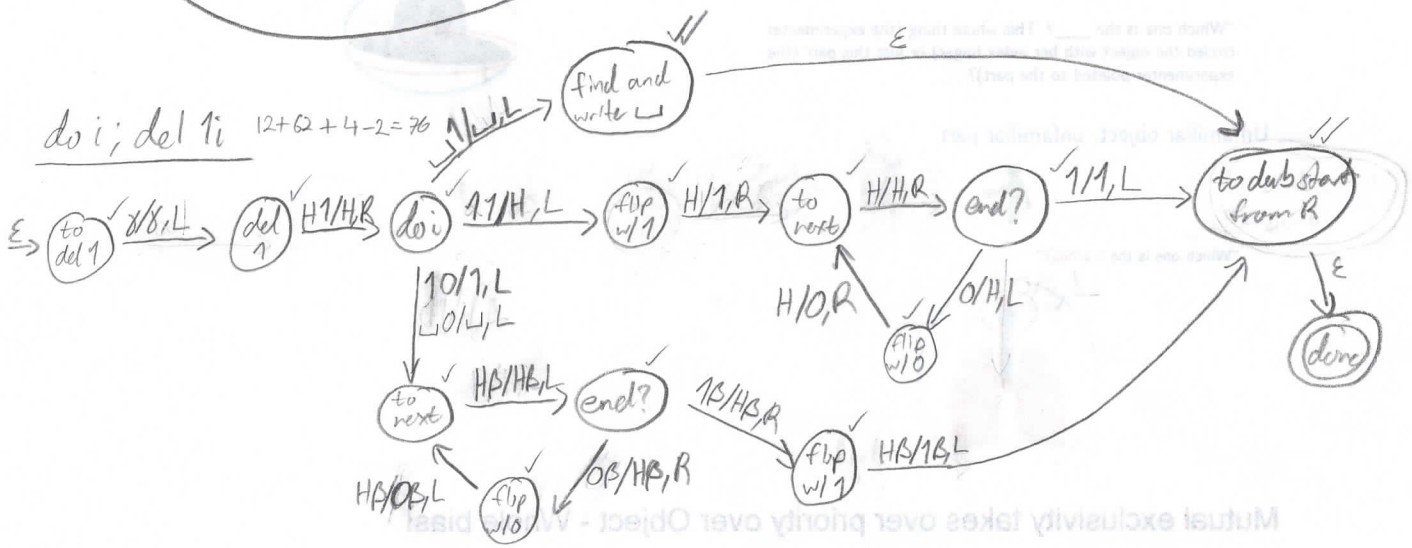
check accept 11 + 7·4 = 39



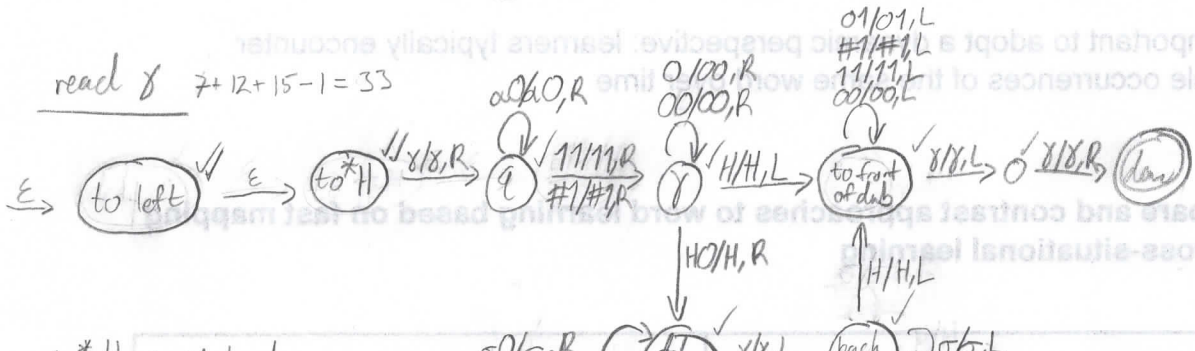
write γ′ 8+4−1=11



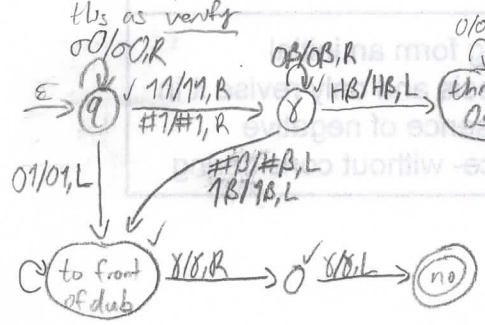
do i; del 1i 12+62+4−2=76



read γ 7+12+15−1=33

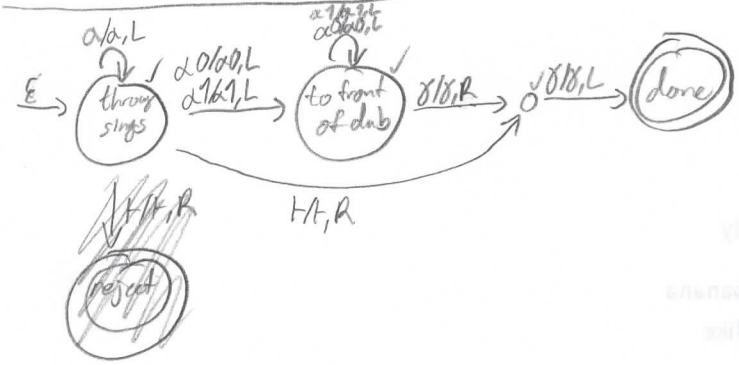


to*H is identical to to H, just with tbs as verify

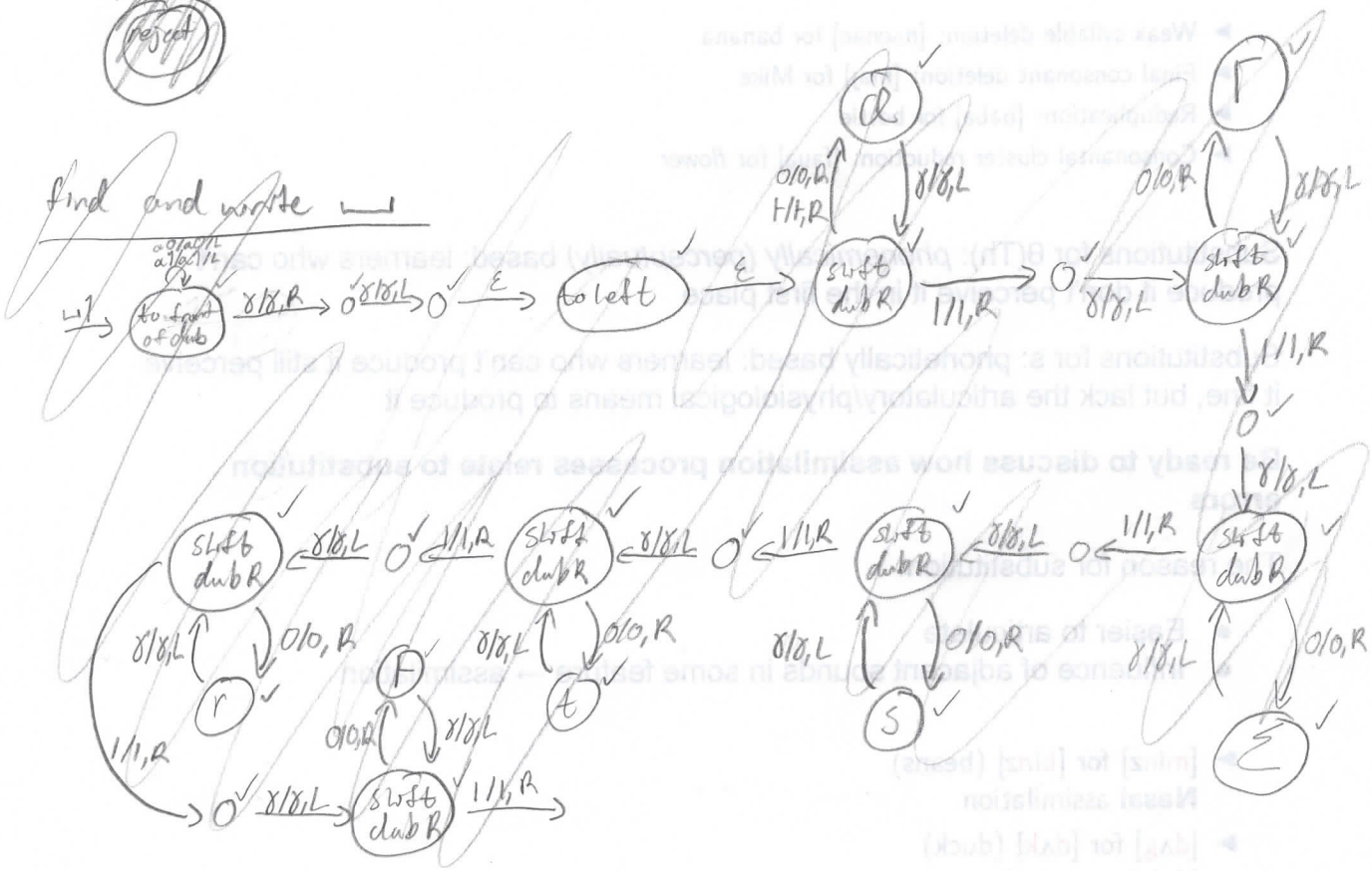


actually, this might be identical, since we don't actually care about i in either.

to dub start from R 4

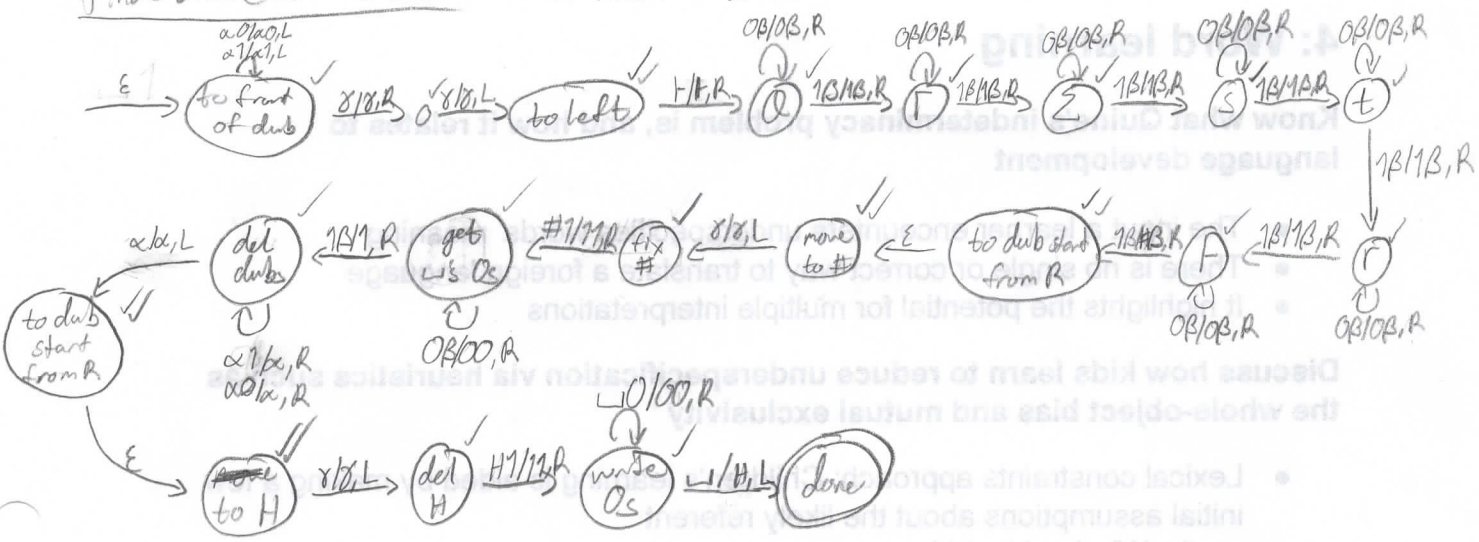


find and write ␣

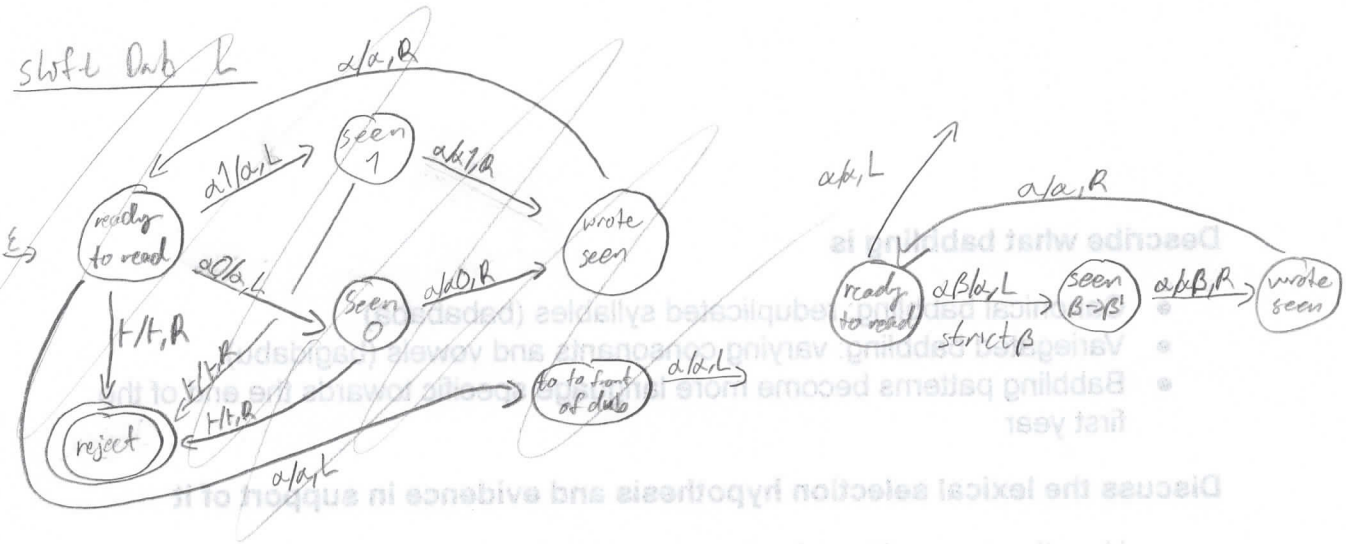


find and write ␣

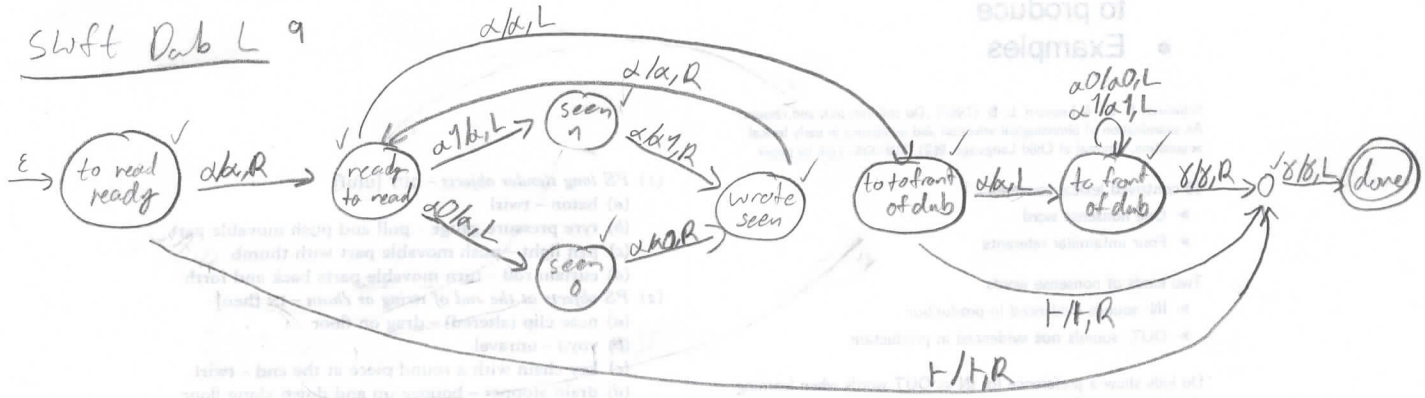
15+12+4·2+14+15−2=62



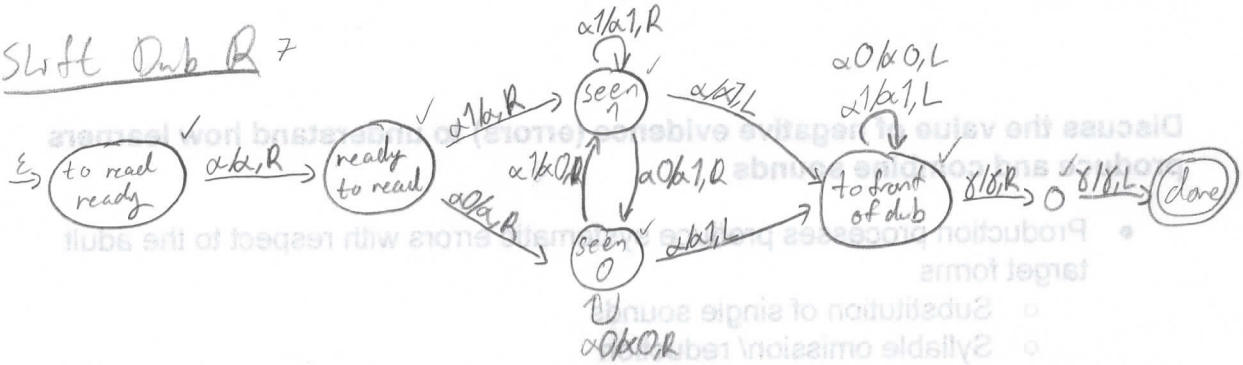
Shift Dub L



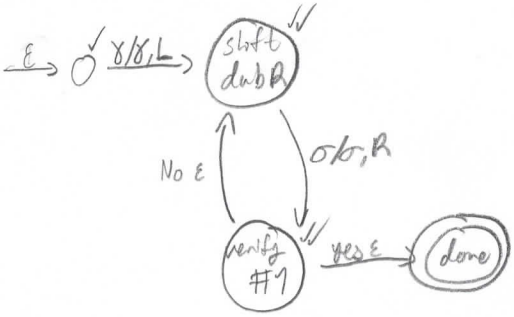
Shift Dub L 9



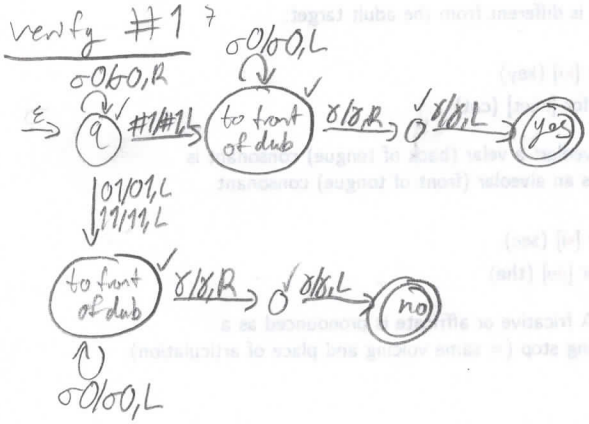
Shift Dub R 7



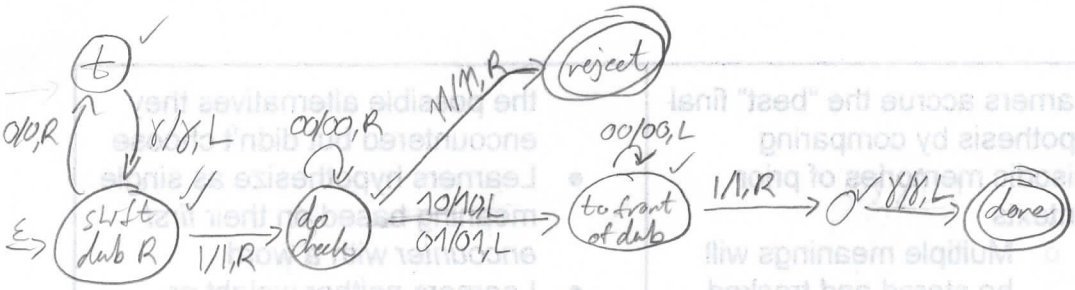
move to # 2+7+7-2=14



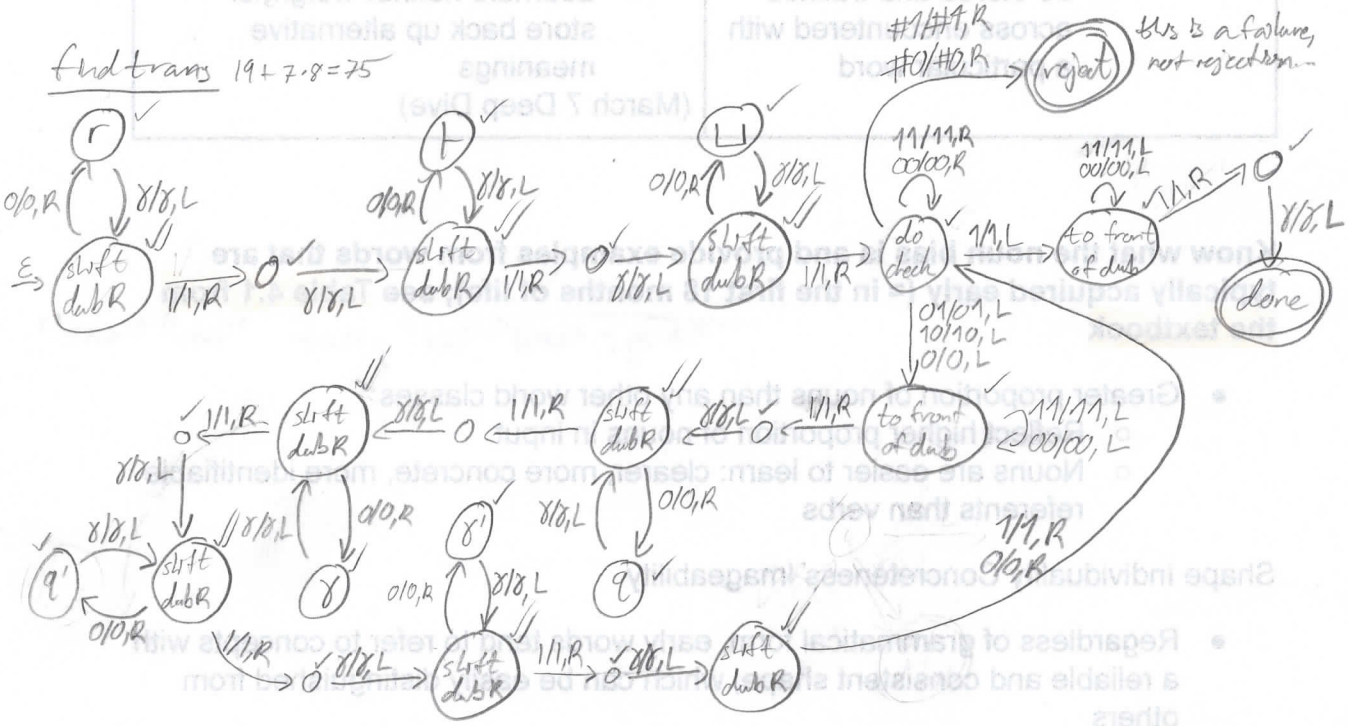
verify #1 7



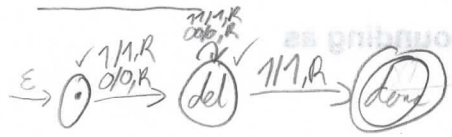
check reject 5+7=12



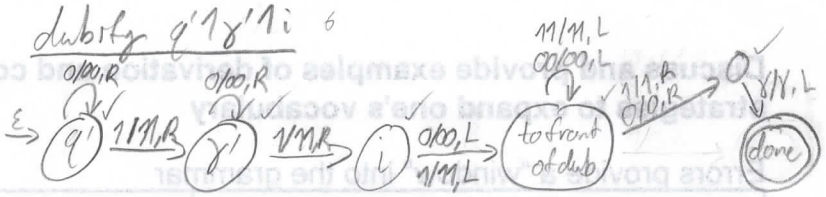
find trans 19+7·8=75



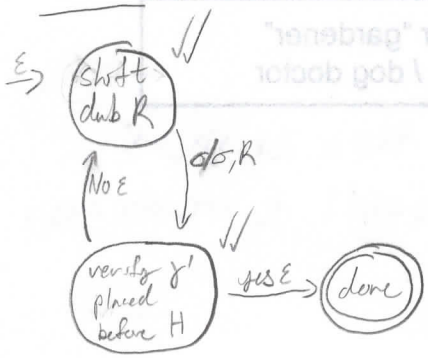
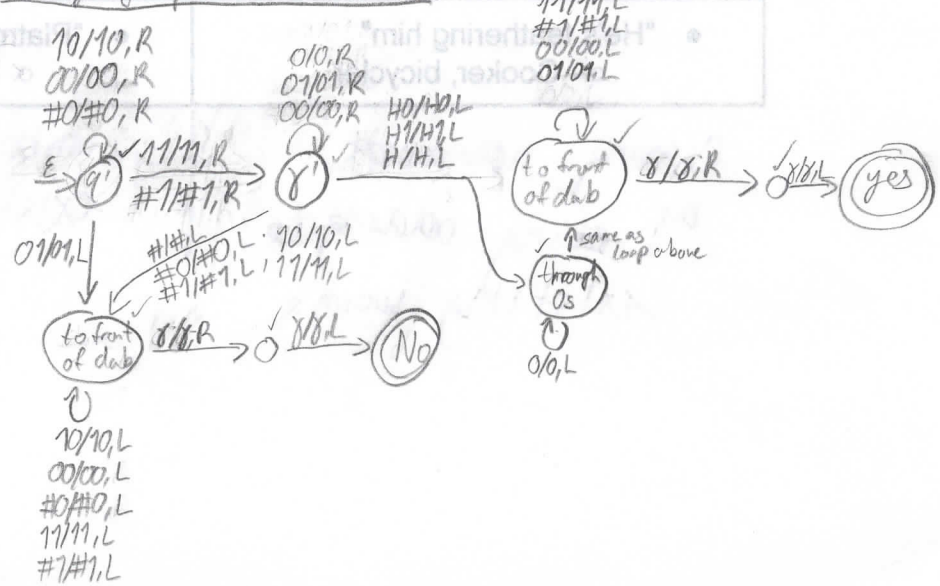
delete dub 3



dubify q'1γ'1i 6



to H 1+7+9-2=15



verify γ' placed before H 9